Harsh Thomare

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Education

University of California, Irvine Dec 2024
Bachelor, Computer Science and Engineering, cum laude, GPA 3.8

Publications

HELIOS: Hierarchical Graph Abstraction for Structure-Aware LLM Decompilation Jan 2026
LAST-X Workshop, NDSS Symposium 2026
Yonatan Gizachew Achamyeleh, **Harsh Thomare**, Mohammad Abdullah Al Faruque. arXiv:2601.14598.

Research Experience

Research Collaborator Apr 2026 - Present
UCLA Security Lab (Prof. Yuan Tian)

- Developed RevBench, a containerized evaluation harness for LLM-based binary decompilers, and analyzed failures using a 14-category defect taxonomy
- Characterized weaknesses that aggregate evaluation scores can obscure, including errors involving structures, loops, and optimized code
- Developing a benchmark to evaluate whether LLM agents can infer hidden program behavior from microarchitectural measurements, using contamination-controlled evaluation

Research Assistant May 2025 - Nov 2025
UC Irvine, AI Cyber-Physical Systems (AICPS) Lab (Prof. Mohammad Abdullah Al Faruque)

- Co-developed HELIOS, a structure-aware approach to LLM-based binary decompilation that represents Ghidra-extracted control-flow information as hierarchical context
- Evaluated HELIOS across six instruction-set architectures and optimization levels from -Oo to -O3; on x86-64 HumanEval-Decompile, compiler feedback yielded 94.9-96.5% compilability and 53.2-55.9% functional correctness across the evaluated models

Undergraduate Researcher Sep 2022 - Mar 2023
UC Irvine, AI Cyber-Physical Systems (AICPS) Lab (Prof. Mohammad Abdullah Al Faruque)

- Built Raspberry Pi-based automation to collect differential pressure-sensor measurements for an acoustic side-channel study
- Evaluated speech-recovery models on recorded sensor data; acknowledged for contributions to "A Fly on the Wall," published at ACSAC 2024

Work Experience

Embedded Software Engineer Oct 2026 - Present
Mainspring Energy *Menlo Park, CA*

- Developing data collection and selection workflows for an ML-based predictive-maintenance system
- Built firmware and over-the-air update infrastructure supporting more than 1,000 deployed devices

Associate Embedded Software Engineer Jul 2024 - Oct 2026
Mainspring Energy *Menlo Park, CA*

- Developed real-time embedded software and automated hardware-testing workflows for deployed energy systems
- Built and validated embedded software across hardware and system-level test environments

Electrical Engineering Intern Jan 2024 - Apr 2024
Second Order Effects

- Built an FPGA data-capture pipeline for high-speed ADCs and verified it with SystemVerilog testbenches
- Wrote a Linux driver to expose captured samples for system-level testing

Projects

Qwen Mega-Kernel: Fused Triton Kernels

- Fused Triton kernels for Qwen and improved achieved memory bandwidth by 30%

IEEE Micromouse Jan 2021 - Dec 2022

- Designed the PCB and built BLE logging for PID tuning

Honors and Awards

- Cum laude, B.S. Computer Science and Engineering, UC Irvine Dec 2024
- UROP Distinguished Research Fellowship, UC Irvine Jan 2023
- Sigma Prize Winner, Mental Health Innovation Challenge Jan 2022

Service

Ambassador Jan 2016 - Present

Akshaya Patra Foundation

- Led a Bay Area sapling-sale initiative, coordinating volunteers and raising funds to feed 5,000 children